Study Report: AI Agents in Action - Agentic AI Engineer

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Family:	Agentic AI Engineer
Level:	Intermediate
Chapters:	7
Source:	Reference: AI Agents in Action - Michael Lanham

Official sources and free libraries

- https://www.manning.com/books/ai-agents-in-action

Summary

Study report for *AI Agents in Action* by Michael Lanham (Intermediate level) mapped to the Agentic AI Engineer role. Agent loops, tool use, planning, and multi-step reasoning.

Chapters

Track: Book-Intro

Chapter 1: About AI Agents in Action [Intermediate]

Chapter focus: About AI Agents in Action *(Level: Intermediate)**

This study report summarizes *AI Agents in Action* by Michael Lanham for the Agentic AI Engineer role. The resource is rated Intermediate level. Agent loops, tool use, planning, and multi-step reasoning. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# AI Agents in Action
# Author: Michael Lanham
# Role: Agentic AI Engineer
# Level: Intermediate
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is
A structured study report based on AI Agents in Action.

When to use it
When learning Agentic AI Engineer skills at Intermediate level.

Where to use it
Agentic AI Engineer training paths and certification prep.

Real use example
A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Intermediate]

Chapter focus: Why This Book Matters for Your Role *(Level: Intermediate)**

Agentic AI Engineer professionals use ideas from AI Agents in Action to solve real workplace problems. Agent loops, tool use, planning, and multi-step reasoning. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

```
Role: Agentic AI Engineer
Book focus: Agent loops, tool use, planning, and multi-step reasoning.
Recommended level: Intermediate
```

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is
Role-aligned learning connects theory to job tasks.

When to use it
During career planning and syllabus design.

Where to use it
Agentic AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example
A teacher assigns this book report before a intermediate skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Intermediate]

Chapter focus: Key Topics Covered *(Level: Intermediate)**

The main topics in AI Agents in Action include practical concepts described as: Agent loops, tool use, planning, and multi-step reasoning. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Agentic AI Engineer jobs today.

Code Reference:

```
- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept
```

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is
Topic maps turn a book into actionable learning objectives.

When to use it
Before reading and while building chapter summaries.

Where to use it
Self-study, flipped classroom, and revision.

Real use example
A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: LangGraph StateGraph Fundamentals [Intermediate]

Chapter focus: LangGraph StateGraph Fundamentals *(Level: Intermediate)**

LangGraph models agents as graphs: nodes mutate shared state; edges route conditionally. Supports cycles, persistence, and interrupts - unlike DAG-only chains.

Code Reference:

```
from langgraph.graph import StateGraph, END

graph = StateGraph(AgentState)
graph.add_node('plan', plan_node)
graph.add_node('act', act_node)
graph.add_edge('plan', 'act')
graph.add_conditional_edges('act', should_continue, {'continue': 'plan', 'end': END})
```

What it shows: conditional_edges implement loop-or-stop logic declaratively.

Topic guide

What it actually is
LangGraph is a framework for durable, stateful agent workflows.

When to use it
Production agents needing checkpoints and branching.

Where to use it
Support automation, coding agents, and ops runbooks.

Real use example
Graph resumes from checkpoint after server restart mid-workflow.

Chapter 5: Applied: MCP: Model Context Protocol [Intermediate]

Chapter focus: MCP: Model Context Protocol *(Level: Intermediate)**

MCP (Anthropic, 2024-2025) standardizes how hosts connect to tools/data via JSON-RPC. Servers expose resources, prompts, and tools - Cursor and Claude Desktop use MCP today.

Code Reference:

```
{
  "jsonrpc": "2.0",
  "method": "tools/call",
  "params": {"name": "query_db", "arguments": {"sql": "SELECT 1"}}
}
```

What it shows: tools/call invokes MCP server capability with typed args.

Topic guide

What it actually is
MCP is an open protocol for portable tool integrations across LLM hosts.

When to use it
Building reusable connectors instead of one-off agent hacks.

Where to use it
IDE assistants, enterprise copilots, and shared data access layers.

Real use example
Railway MCP server deploys services - agent uses same tool in Cursor and internal bot.

Chapter 6: Applied: Building an MCP Server [Intermediate]

Chapter focus: Building an MCP Server *(Level: Intermediate)**

Python MCP SDK: define @server.list_tools and @server.call_tool handlers. Run stdio for local hosts or SSE for remote. Validate inputs; never expose raw shell.

Code Reference:

```
from mcp.server import Server
server = Server('docs-search')

@server.list_tools()
async def list_tools():
    return [Tool(name='search', description='Search internal wiki', inputSchema={...})]
```

What it shows: list_tools advertises capabilities; call_tool executes with validation.

Topic guide

What it actually is
Custom MCP servers package domain tools for any compatible host.

When to use it
Shared internal tools across Claude, Cursor, and custom agents.

Where to use it
Wiki search, Jira, and data catalog integrations.

Real use example
Docs MCP server powers both IDE and Slack agent with one codebase.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Intermediate]

Chapter focus: Study Plan and Practice *(Level: Intermediate)**

Finish AI Agents in Action with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Agentic AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is
Structured study plans improve retention and portfolio outcomes.

When to use it
After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a intermediate project screenshot.